Sound

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March 9, 2026

This article was prepared by the author to serve as a draft for TNSCERT class 8 chapter on the concept of Sound for TNSCERT during the the Tamil Nadu Curriculum Framing(TNCF) workshop that was held in Asha Nivas, Chennai from 13-11-2025 to 15-11-2025, and 09-03-2026 to 10-03-2026.

The right side margin of each page contains notes for teachers, and prompts to activities that the teacher can initiate in the classroom.

Every day we experience many natural and human-made sounds around us: the school bell ringing, raindrops falling on the ground, a dog barking, or a phone ringing. But how are these sounds produced? How do they propagate[1] from one place to another? Why do different sounds have different characteristics? In this chapter, we will explore how sound is produced and how it propagates. We will also learn to think of sound as a form of energy and understand it as a wave.

Write down all the sounds that are familiar to you.

[1] Here the word 'propagate' is used instead of simpler and familiar word like 'travel' to avoid misconceptions that students create, e.g. The word travel suggests that 'sound' is a moving object. Words are chosen to ensure that students develop a scientifically accurate mental model of the concept.

How is Sound Produced?

Activity: Vibrating Scale

Materials required:

1. Geometry box including scale
2. Rubber bands
3. Rough paper
4. Scissors

Hang the scale over the edge of the desk so that part of it sticks out. Push the free end of the scale downwards and then let it go.

Figure 1: Step 1. Push the free end of the scale downwards.
Image credit: Ingrid Sulston, licensed under CC BY-NC-SA 4.0

Observe: Do you hear a sound? Do you see the scale move up and down rapidly? What kind of motion is the scale showing?

Musical Geometry Box

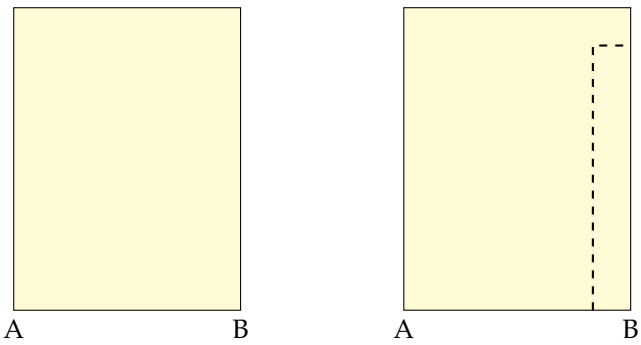
Empty your geometry box. Stretch a rubber band tightly around the box. Pluck or poke the rubber band.

Observe: Do you hear a sound? What part of the setup is moving?

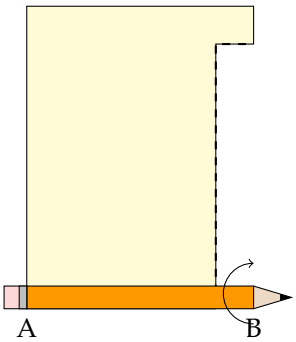
Paper whistle

Create your own paper whistle as shown in figure . Suck air in through the open end.

Figure 2: Step 2. Let the pushed end go.
Image credit: Ingrid Sulston, licensed under CC BY-NC-SA 4.0



Step 1: Take a piece of paper. Step 2: Cut along the dashed lines



Step 3: Place a pencil along AB and roll the paper into a cylinder

Step 4: Remove the pencil. Fold the flap to close one side of the cylinder

Figure 3: Instructions for creating your own sound maker.

Optional Activity: Tuning Fork

This activity can be done if a tuning fork is available.

Strike a prong of a tuning fork gently with a rubber hammer to make it vibrate. Bring the vibrating tuning fork close to your ear. Can you hear a sound?

Gently touch the prongs of the vibrating tuning fork with your finger. What do you feel? Strike the tuning fork again. Lightly touch the vibrating prongs to the surface of water in a bowl. Do you see ripples forming on the surface of the water?

Figure 4: Tuning fork

DISCUSSION: In each case, some part of the object moves back and forth rapidly. This type of motion is called **vibration**.

SOUND IS PRODUCED WHEN OBJECTS VIBRATE.

The vibrating part of an object can cause nearby objects, such as air or water, to vibrate and form ripples. The vibrations propagate to us as sound.

Create a table as shown in Table listing the object that produces sound and the part of the object that vibrates.

Did you know? Many quartz clocks and watches keep time using a small quartz crystal. This crystal is shaped like a tiny tuning fork.

Sl. No.	Object that produces sound	Part of the object that vibrates
1	Vibrating scale	Tip of the scale
2	Musical geometry box	⋮
3	⋮	⋮
⋮	⋮	⋮

Table 1: Observation table

How Sound Propagates?

Activity

Materials required:

1. Mobile phone
2. Airtight container

Switch on music on the mobile phone and place it inside the airtight container. Close the lid of the container tightly.

OBSERVE: What happens to the sound from the mobile phone?

DISCUSSION: Sound propagates through air and reaches our ears. Sound cannot propagate through vacuum.

Listening through wood or metal

Place your ear on the desk. Ask your benchmate to gently tap the desk with a pencil a short distance away from your ear. Now compare listening with your ear in the air and listening with your ear touching the table.

OBSERVE: The sound is usually louder or clearer through the table.

DISCUSSION: Sound can propagate through solids, not just through air.

Sound needs a medium (such as air, water, or solids) to propagate. A material through which sound propagates is called a **medium**.

Does air travel with the sound?

Materials required:

1. A piece of tissue paper or a candle
2. A sound-producing object (such as a bell, or a plate and spoons)

Place the candle (or hang the tissue paper) about 20-30 cm away from the sound source. Produce a sound.

OBSERVE: Watch the flame or tissue carefully. It usually does not move much, or moves only slightly. If air were travelling from the sound source to your ear like wind, what should happen to the flame or tissue?

DISCUSSION: In sound, the air particles vibrate back and forth in place. Each particle pushes the next one, passing the vibration along. The vibration propagates through the medium, not the particles themselves moving across the room In this activity, the air itself is not moving across the room.

Sound Waves

We have seen that sound is produced when objects vibrate. These vibrations make the surrounding air vibrate. Each air particle pushes the next one, passing the vibration along, like a chain of pushes. In this way, the vibration propagates through the air from the sound source to our ears. When vibrations move through a medium like this, they form a wave. Sound propagates through air as a sound wave.

Compressions and Rarefactions

When sound propagates through air, the air particles move back and forth around their positions. As they move, they sometimes come closer together and sometimes move farther apart. The regions where air particles are closely packed together are called **compressions**. The regions where air particles are farther apart are called **rarefactions**.

As a sound wave propagates through air, compressions and rarefactions move forward through the air. This pattern of compressions and rarefactions carries the sound from the source to our ears.

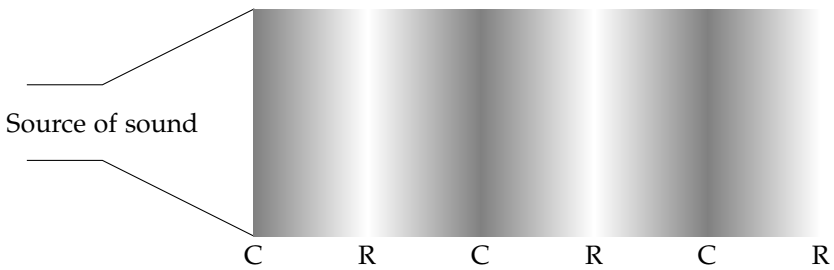


Figure 5: Sound propagates as density variations in the medium. C: Compression, R: Rarefaction

Optional Activity: Slinky

Take a slinky or a long spring. Ask your classmate to hold one end while you hold the other end. Stretch the slinky gently between you. Now give the slinky a quick push toward your classmate.

Figure 6: Long spring also known as slinky.

OBSERVE: What happens when you push and pull the slinky back and forth along its length? Tie a ribbon somewhere in the middle of the slinky. Watch the ribbon carefully. You will notice that the ribbon moves back and forth, parallel to the direction in which the disturbance propagates.

This is how sound waves propagate: molecules bump the next molecules along, forming a wave of vibrations. These are called **longitudinal waves**, and is how sound propagates through solids, liquids and gases.

How Fast Does Sound Propagate?

Do we hear sound instantly after it is produced? Think about what happens during lightning and thunder. When lightning strikes, we see a flash of light and hear thunder.

Figure 7: Thunderstorm in Tiruchirappalli, Photo by Amol Mande

OBSERVE: Do you see the lightning and hear the thunder at the same time? Or do they happen one after the other? Which do you notice first?

Usually, we see the lightning first and hear the thunder later.

Now think about what happens when crackers burst.

These observations show that sound takes time to propagate from the source to our ears. Light travels much faster than sound, so we see the flash before we hear the sound.

In air, sound propagates at a speed of about 340 metres per second (at room temperature). Does sound propagate at the same speed through all mediums?

Sound does not propagate at the same speed in all materials. The speed of sound depends on the medium through which it propagates. Sound propagates fastest in solids, slower in liquids, and slowest in gases. For example, sound propagates faster in metal or wood than in air. This is because the particles in solids are closer together, allowing vibrations to pass more quickly from one particle to the next.

How does temperature affect the speed of sound?

As the temperature increases, the speed of sound increases. This happens because when the temperature is higher, the particles of the medium move faster. As a result, they pass the vibrations to neighbouring particles more quickly, allowing sound to propagate faster. For example, sound travels faster in warm air than in cold air.

Activity: Vibrating Scale

Place a scale on the edge of a desk so that part of it hangs over the edge. Hold the scale firmly on the desk and pluck the free end so that it vibrates and produces a sound. Optionally, you may use a phone to record a video of the scale in slow-motion setting.

OBSERVE: How does the sound and speed of vibration change as you change the length of the overhanging edge? How fast does the scale vibrate?

You may notice that when the scale is longer, it vibrates more slowly and the sound is lower and when the scale is shorter, it vibrates faster and the sound is higher. If you have a slow motion

recording, you can replay the video and count how many times the scale moves back and forth in 5 second and divide the number by 5 to find out how many times it moved back and forth in one second.

When the scale vibrates, it also makes the surrounding air vibrate. As the scale moves back and forth, it pushes and pulls the nearby air. This creates regions where the air is slightly compressed and regions where it is spread out. These compressions and rarefactions propagate through the air as sound waves. When the sound waves reach our ears, they cause the eardrum to vibrate, which allows us to hear the sound.

DISCUSSION: the number of vibrations that occur in one second is called the **frequency** of the vibration. Frequency is measured in hertz (Hz) and usually represented by the Greek letter ν (nu). Frequency tells us how often an event occurs in a unit of time.

EXAMPLE: if you beat a drum 10 times in one second, the frequency of your beating is 10 Hz.

Sounds with higher frequency have a higher pitch and sounds with lower frequency have a lower pitch. A mosquito makes a high-pitched sound. A drum makes a low-pitched sound. Which one is vibrating faster?

When the scale vibrates, it sets the surrounding air into vibration. As the scale moves back and forth, it creates regions where the air is slightly compressed and regions where the air is slightly spread out. These compressions and rarefactions move through the air as sound waves. When they reach our ears, they cause the eardrum to vibrate, allowing us to hear the sound.

Kavin and Trisha are standing at different distances from the lightning. Both hear the thunder. Will the pitch of the thunder be different for Kavin and Trisha? Will the loudness of the thunder be different?

You may also have noticed something else during the activity. When the scale was plucked gently, the sound was soft. When the scale was plucked harder, the sound was louder. If you watch the scale carefully, you will see that when it is plucked harder, it moves farther away from its resting position during each vibration.

The size of this vibration is called the **amplitude**. Amplitude tells us how large the vibration is. A vibration with greater amplitude produces a louder sound, while a vibration with smaller amplitude produces a softer sound. Amplitude affects loudness, not pitch.

Two sounds have the same pitch, but one is louder. What must be different?

Hearing loss and hearing aids

Very loud sounds can damage the inner ear and lead to hearing loss. Some people are born with conditions that affect their ability to hear.

Hearing aids are devices that help such people hear better. They work by amplifying sound, that is, by increasing the amplitude of the sound waves.

very loud sounds can damage the inner ear this leads to hearing loss some people have congenital condition hearing aids amplify sound (increase amplitude)

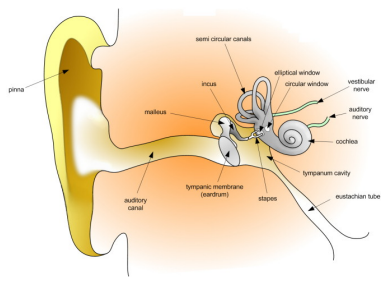
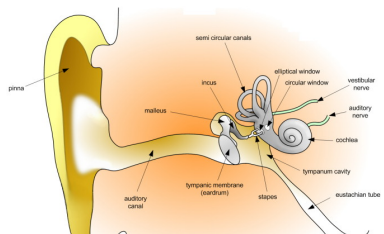


Figure 8: Diagram of the auditory system of the human ear. Illustration by Dan Pickard.



Musical instruments

How varying components of the instruments can vary the frequency

Ultrasound

humans hear roughly 20 Hz – 20,000 Hz sounds with higher frequency are ultrasound

medical imaging detecting cracks in materials animal communication (bats, dolphins)

Activity: Slinky

Take a slinky or a long spring. Ask your classmate to hold one end while you hold the other end. Stretch the slinky gently between you. Now give the slinky a quick push toward your classmate.

OBSERVE: a compression moves along the slinky. You will see regions where the coils are close together and regions where the coils are spread apart. The region where the coils are close together is called a compression. The region where the coils are farther apart is called a rarefaction. This is similar to what happens in sound waves in air.

The distance between two neighbouring compressions (or two neighbouring rarefactions) is called the **wavelength**. Wavelength is measured in the units of distance. Remember, metre is the SI unit of distance.

Repeat the activity, but now create compressions more quickly.

OBSERVE: The compressions become closer together, so the wavelength becomes shorter. If vibrations happen more often each second (higher frequency), the compressions are packed closer together and the wavelength becomes shorter.

When sound travels through a medium, regions of compression and rarefaction move through the medium. The distance between compressions is the wavelength. The speed of a sound wave depends on both its frequency and its wavelength.

Echo

sound travels at a finite speed sound can reflect from surfaces An echo is heard when reflected sound reaches our ears after a short delay.

Echolocation

Kanna muchale aata bats dolphins some visually impaired people They produce sounds and detect the reflected sound to locate objects.

$$\text{Speed} = \text{Frequency} \times \text{Wavelength} \tag{1}$$

If the thunder sounds quieter when you are farther away, does the thunder have less energy when it reaches you?

Noise Pollution

traffic noise loudspeakers industrial noise How can we reduce noise pollution in cities?

Sound is one example of a mechanical wave that is longitudinal. There is also another type of wave, called a transverse wave. In a transverse wave particles do not oscillate along the direction of wave propagation but oscillate up and down about their mean position as the wave travels. Thus, a transverse wave is the one in which the individual particles of the medium move about their mean positions in a direction perpendicular to the direction of wave propagation. When we drop a pebble in a pond, the waves you see on the water surface is an example of transverse wave. Light is a transverse wave but for light, the oscillations are not of the medium particles or their pressure or density — it is not a mechanical wave.

References

Haim Eshach, Tzu-Chiang Lin, and Chin-Chung Tsai. Taiwanese middle school students' materialistic concepts of sound. *Phys. Rev. Phys. Educ. Res.*, 12:010119, Mar 2016. DOI: 10.1103/PhysRevPhysEducRes.12.010119. URL `https://link.aps.org/doi/10.1103/PhysRevPhysEducRes.12.010119`.